

Data Preprocessing

Data Mining: Concepts and Techniques
Chapter 3

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Data Preprocessing: An Overview

- Data preprocessing forms the foundation of any successful machine learning project, as it directly impacts the quality and effectiveness of the models we build.
- Preparing our data correctly ensures that the models can extract meaningful patterns and insights, leading to accurate predictions and valuable insights.
- Raw data is often messy, incomplete, and inconsistent. It may contain noise, irrelevant information, and outliers that can negatively impact the performance of machine learning models. Data preprocessing aims to clean, transform, and restructure the data so that it becomes suitable for model training.

Data Quality: Why Preprocess the Data?

Measures for data quality: A multidimensional view

- Accuracy: correct or wrong, accurate or not
- Completeness: not recorded, unavailable, ...
- Consistency: some modified but some not, dangling, ...
- Timeliness: timely update?
- Believability: how trustable the data are correct?
- Interpretability: how easily the data can be understood?

Major Tasks in Data Preprocessing

- Data cleaning
 - Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- Data integration
 - Integration of multiple databases, data cubes, or files
- Data reduction
 - Dimensionality reduction
 - Numerosity reduction
 - Data compression
- Data transformation and data discretization
 - Normalization
 - Concept hierarchy generation

Data Cleaning

- Data in the Real World Is Dirty: Lots of potentially incorrect data, e.g., instrument faulty, human or computer error, transmission error
 - incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
 - e.g., Occupation="" (missing data)
 - noisy: containing noise, errors, or outliers
 - e.g., Salary="-10" (an error)
 - inconsistent: containing discrepancies in codes or names, e.g.,
 - Age="42", Birthday="03/07/2010"
 - Was rating "1, 2, 3", now rating "A, B, C"
 - discrepancy between duplicate records
 - Intentional (e.g., disguised missing data)
 - Jan. 1 as everyone's birthday?

Incomplete (Missing) Data

- Data is not always available
 - E.g., many tuples have no recorded value for several attributes, such as customer income in sales data
- Missing data may be due to
 - equipment malfunction
 - inconsistent with other recorded data and thus deleted
 - data not entered due to misunderstanding
 - certain data may not be considered important at the time of entry
 - not register history or changes of the data
- Missing data may need to be inferred

How to Handle Missing Data?

- Ignore the tuple: usually done when class label is missing (when doing classification)—not effective when the % of missing values per attribute varies considerably
- Fill in the missing value manually: tedious + infeasible?
- Fill in it automatically with
 - a global constant : e.g., “unknown”, a new class?!
 - the attribute mean
 - the attribute mean for all samples belonging to the same class: smarter
 - the most probable value: inference-based such as Bayesian formula or decision tree

Noisy Data

- Noise: random error or variance in a measured variable
- Incorrect attribute values may be due to
 - Faulty data collection instruments
 - Data entry problems
 - Data transmission problems
 - Technology limitation
 - Inconsistency in naming convention
- Other data problems which require data cleaning
 - Duplicate records
 - Incomplete data
 - Inconsistent data

How to Handle Noisy Data?

- Binning
 - First sort data and partition into (equal-frequency) bins
 - Then one can smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.
- Regression
 - Smooth by fitting the data into regression functions
- Clustering
 - Detect and remove outliers
- Combined computer and human inspection
 - Detect suspicious values and check by human (e.g., deal with possible outliers)

Data Reduction Strategies

- Data reduction: Obtain a reduced representation of the data set that is much smaller in volume but yet produces the same (or almost the same) analytical results
- Why data reduction? — A database/data warehouse may store terabytes of data. Complex data analysis may take a very long time to run on the complete data set.
- Data reduction strategies
 - Dimensionality reduction, e.g., remove unimportant attributes
 - Numerosity reduction (some simply call it: Data Reduction)
 - Clustering, sampling
- Data compression

Data Normalization

- Min-max normalization: to $[new_min_A, new_max_A]$

$$v' = \frac{v - min_A}{max_A - min_A} (new_max_A - new_min_A) + new_min_A$$

- Ex. Let income range \$12,000 to \$98,000 normalized to $[0.0, 1.0]$. Then \$73,000 is mapped to $\frac{73,600 - 12,000}{98,000 - 12,000} (1.0 - 0) + 0 = 0.716$

- Z-score normalization (μ : mean, σ : standard deviation): $v' = \frac{v - \mu_A}{\sigma_A}$

- Ex. Let $\mu = 54,000$, $\sigma = 16,000$. Then $\frac{73,600 - 54,000}{16,000} = 1.225$

- Normalization by decimal scaling

Summary

- Data quality: accuracy, completeness, consistency, timeliness, believability, interpretability
- Data cleaning: e.g. missing/noisy values, outliers
- Data integration from multiple sources:
 - Entity identification problem
 - Remove redundancies
 - Detect inconsistencies
- Data reduction
 - Dimensionality reduction
 - Numerosity reduction
 - Data compression
- Data transformation and data discretization
 - Normalization
 - Concept hierarchy generation

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